

Observation-Guided Physics-Informed Neural Network (OG-PINN): Application to Subsurface Ocean Temperature and Salinity Structure Reconstruction

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Observation-Guided Physics-Informed Neural Network (OG-PINN): Application to Subsurface Ocean Temperature and Salinity Structure Reconstruction

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Abstract

Marine observations are essential for climate forecasting, weather prediction, and Earth system dynamics research. However, due to cost and technological constraints, sub-surface ocean observations suffer from insufficient spatial coverage and poor temporal resolution. While data assimilation products provide three-dimensional fields, they often lag behind real-time requirements. To address this, we propose an Observation-Guided Physics-Informed Neural Network (OG-PINN), which efficiently reconstructs the three-dimensional temperature, salinity, and density structure of the ocean interior using surface observations. By embedding the seawater equation of state as a physical constraint within the network architecture, OG-PINN ensures thermodynamic consistency among temperature, salinity, and density. Crucially, by incorporating observational data into the physical loss, OG-PINN effectively resolves the optimization conflict between data and physical constraints inherent in standard Physics-Informed Neural Networks (PINNs).

OG-PINN outperforms U-Net and standard PINNs in reconstructing subsurface temperature-salinity (T-S) structures by adaptively adjusting the weight of the physical loss function to balance data-fitting accuracy and physical constraint. Results demonstrate that OG-PINN reduces the root mean square error (RMSE) of tropical mean subsurface temperature anomalies (STA) and subsurface salinity anomalies (SSA) by up to 5%, while spatially averaged correlation coefficients between reconstructed fields and ground truth improve by up to 7% over standard PINNs. Notably, in dynamically complex upwelling regions, OG-PINN mitigates adverse effects from idealized physical constraints by dynamically reducing physical loss weight, whereas standard PINNs suffer from spurious convergence in physical losses, failing to achieve reasonable weight adjustment. This study not only validates the critical role of observation-guided physical constraints in enhancing both accuracy and

physical consistency of sub-surface variable reconstruction, but also provides high-quality, timely initial fields for temperature and salinity to climate models. These advances hold significant scientific and practical implications for improving the predictability of key tropical climate modes such as El Niño-Southern Oscillation (ENSO).

Keywords: Deep-Learning, Physics-Informed Neural Networks (PINN), Observation-Guided PINN, Ocean Temperature and Salinity Reconstruction

1. Introduction

The state of the ocean subsurface is fundamental to understanding oceanic dynamical processes, predicting climate variability, and supporting marine engineering applications. Key oceanic phenomena, such as mesoscale eddies, internal wave propagation, and thermocline variability, occur primarily in the ocean interior. Moreover, as the largest heat and carbon reservoir in the Earth system, the subsurface distribution of temperature and salinity directly modulates the evolution of major climate modes, including the El Niño–Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD) (Behera et al., 2006; Deshpande et al., 2014; Fedorov, 2010; Duan & Wei, 2013; Feng et al., 2017). Consequently, achieving highly accurate, high spatiotemporal-resolution three-dimensional (3D) reconstructions of the ocean interior holds irreplaceable scientific significance for elucidating the mechanisms of climate variability and improving predictive capability.

However, observing the ocean subsurface remains a formidable challenge. Although observational systems, including in situ observation, buoy networks, and satellite remote sensing, have grown increasingly diverse. Notably, the global Argo float array and multi-source satellite data have significantly improved spatiotemporal coverage (Morrow et al., 2019; Roemmich et al., 2009). Nevertheless, intrinsic limitations hinder their effectiveness for ocean dynamics research and climate prediction: in situ and buoy observations are spatially sparse and temporally discontinuous, while satellite remote sensing only captures sea surface information, as electromagnetic signals cannot penetrate below the surface (Li et al., 2021; Luo et al., 2025). This “data-rich at the surface, data-poor in the interior” paradigm has long impeded the construction of continuous, complete 3D ocean state fields.

Fortunately, the ocean surface and subsurface are dynamically coupled. Governed by geostrophic balance and baroclinic mode propagation, many subsurface variations, such as thermocline variability and density anomalies, induce measurable signals at the sea surface, including sea surface height anomalies (SSHA) and sea surface temperature anomalies (SSTA) (Kobashi & Xie, 2011; Willis et al., 2004). This physical linkage provides a robust theoretical foundation for inferring the 3D subsurface structure from abundant surface observations.

On this basis, existing inversion approaches can be broadly classified into two

categories: dynamical and statistical methods. Dynamical methods solve the governing equations of geophysical fluid dynamics and employ data assimilation techniques (e.g., HYCOM, GLORYS reanalysis systems) to reconstruct 3D ocean fields (Lellouche et al., 2018; Zhang et al., 2009). While physically consistent, these models are computationally expensive, lack timeliness, and subject to biases introduced by parameterization schemes. In contrast, statistical methods rely on empirical relationships between surface and subsurface variables. Traditional approaches, such as multiple linear regression or EOF-based reconstruction, are simple but struggle to capture complex nonlinear dynamics (Su et al., 2022; Zhang et al., 2025). In recent years, deep learning models, with their powerful nonlinear fitting and automatic feature extraction capabilities, have demonstrated significant advantages in ocean data reconstruction (Tian et al., 2022). In particular, convolutional neural networks (CNNs) are effective at capturing spatial correlations. For instance, Su et al. (2022) employed ConvLSTM to merge multi-source satellite and Argo data, producing a global subsurface temperature product spanning 1993–2020; Meng et al. (2021) reconstructed 3D temperature and salinity fields from 0–2000 m at $1/4^\circ$ resolution using only satellite surface data using a CNN architecture.

Despite their success, purely data-driven models often lack physical consistency and interpretability. To overcome these challenges, Raissi et al. (2019) introduced Physics-Informed Neural Networks (PINNs), which incorporate residuals of governing partial differential equations (PDEs) as regularization terms in the loss function, achieving joint optimization of data fidelity and physical constraints (Raissi et al., 2019). This framework has inspired a new paradigm for integrating physical priors with data-driven modeling. In oceanography, specialized architectures have been developed accordingly. For example, Liu et al. (2024) proposed 3D-EddyNet, incorporating eddy dynamical priors to reconstruct 3D thermohaline structures, while Zhang et al. (2025) developed a Physics-Constrained Dual-Link Operator Learning Network (PC-DLON) that integrates satellite-derived surface density anomalies with in situ data under quasi-geostrophic theory to reconstruct 3D density fields.

Nevertheless, the implementation of PINNs presents several practical challenges. The total loss comprises data loss, physical loss (from PDE residuals), and boundary condition terms, which often can span several orders of magnitude. This imbalance can bias optimization toward dominant terms, weakening the effectiveness of physical constraints (Beucler et al., 2021; Krishnapriyan et al., 2021; Lu et al., 2021). Although adaptive weighting strategies, such as those based on neural tangent kernels or GradNorm, have been proposed to mitigate this issue (Chen et al., 2018; Jacot et al., 2021), their performance remains unstable. More fundamentally, observational errors and idealized assumptions in physical models can introduce inherent conflicts between data and physics objectives, resulting in training instability or unphysical solutions.

To address these limitations, we propose an Observation-Guided Physics-Informed Neural Network (OG-PINN) for reconstructing 3D ocean temperature and salinity fields from sea surface states. Based on a U-Net backbone, OG-PINN features multi-variable output branches and embeds the seawater equation of state as a physical constraint, ensuring thermodynamic self-consistency among temperature, salinity, and

density. This design harmonizes the optimization trajectories of the data and physical constraints during training, effectively mitigating their competition and enhancing both convergence stability and physical plausibility.

The main contributions of this study are as follows:(1) We develop a OG-PINN architecture based on U-Net that integrates the seawater equation of state to jointly reconstruct tropical subsurface (0–200 m) temperature anomalies (STA), salinity anomalies (SSA), and density anomalies (SDA) from surface states; (2) We innovatively enhance PINN by introducing an observation-guided mechanism with adaptive weighting, which effectively mitigates the optimization conflict between data and physical constraints, significantly improving convergence stability and physical consistency; (3) By analyzing the spatial variability in model performance integrated by physical constraints, we reveal persistent challenges in regions dominated by strong dynamical processes and elucidate the underlying mechanisms by which OG-PINN achieves superior performance in these critical zones.

The remainder of this paper is organized as follows: Section 2 introduces the model architecture, theoretical foundations, and datasets; Section 3 details the training strategy and experimental setup; Section 4 presents and discusses the results; and Section 5 concludes with a summary and future research directions.

2. Data and Methods

2.1 Neural Network Architecture

In this study, we employ the classical U-Net deep neural network as the base model. U-Net was originally proposed by Ronneberger et al. (2015) to address medical image segmentation tasks, and it has since become widely adopted due to its simplicity, efficiency, and robustness in small-data scenarios (Ronneberger et al., 2015). We further extend U-Net to develop a multi-task framework incorporating the Convolutional Block Attention Module (CBAM) (Woo et al., 2018) to enhance feature learning (Fig. 1). Unlike the traditional U-Net, our network adopts a shared encoder and three independent decoders designed for predicting STA, SSA, and SDA, respectively. This architecture not only enables more effective modeling of the complex interrelationships among ocean state variables but also preserves the distinct physical characteristics of each variable.

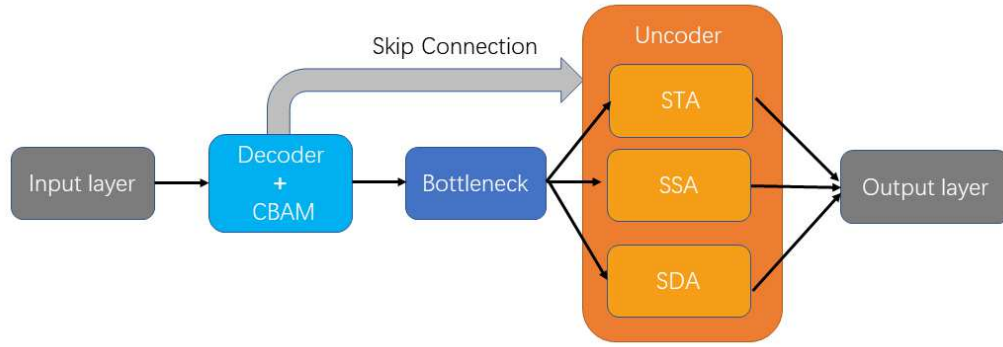


Figure 1 Multi-Task Output U-Net Architecture Incorporating CBAM

Specifically, the independent decoder design offers multiple advantages. First, by assigning dedicated upsampling and feature-fusion modules to each prediction task, the model can optimize its dedicated feature decoder, thereby mitigating gradient competition caused by parameter sharing (Bourgain et al., 2013; Cravatte et al., 2009). Second, the independent decoders enhance the model's ability to represent the vertical structure and spatial heterogeneity of different variables.

To further improve local feature extraction, we introduce the CBAM attention mechanism into the encoder. This mechanism adaptively strengthens key feature representations through both channel and spatial attention, and outputs are added back to the original features in a residual form (Woo et al., 2018), ensuring efficient gradient propagation during training.

The main hyperparameter configurations are as follows. The model input consists of three-channel surface variables—sea surface temperature anomaly (SSTA), sea surface salinity anomaly (SSSA), and sea surface height anomaly (SSHA). The encoder channel numbers are set to [64, 128, 256, 512, 1024]. Each decoder outputs 13 vertical layers, which are concatenated into a 39-channel output corresponding to STA, SSA, and SDA, respectively. The backbone U-Net is optimized using the Adam optimizer combined with a ReduceLROnPlateau learning rate scheduler (Smith, 2017). When the validation loss decreases by less than 10^{-4} for 10 consecutive epochs, the learning rate is multiplied by a decay factor of 0.5, with a minimum threshold of 10^{-8} . This adaptive fine-tuning strategy improves both convergence and generalization performance (Bengio, 2012; Byadarhaly et al., 2012; Smith, 2017). The learning rate for physical constraint weights is fixed at 10^{-5} . The batch size is set to 16, and the total number of training epochs is 300.

The CBAM spatial attention module uses a 7×7 convolution kernel, with bilinear upsampling, Batch Normalization, and ReLU activation functions applied (Glorot et al., 2011; Ioffe & Szegedy, 2015; Long et al., 2015).

This architecture efficiently extracts cross-variable shared representations through the shared encoder while the independent decoders preserve the unique physical structures and spatial patterns of each oceanic state variable. Consequently, it substantially enhances multivariate reconstruction accuracy while maintaining physical consistency.

To address undefined physical quantities (e.g., temperature and salinity over land), we construct land masks during forward propagation and force predictions over land grids to zero. This ensures loss computation is restricted to valid ocean grids, improving training stability.

2.2 Construction of the PINN Loss Function

2.2.1 Physics-Informed Neural Networks (PINN)

Following the framework proposed by Raissi et al. (2019), the PINN model can be trained by minimizing the following composite loss function to achieve the integration of observational data with the physical constraints (Raissi et al., 2019):

$$\mathcal{L}(\theta) = (1 - \lambda)\mathcal{L}_{MSE}(\theta) + \lambda\mathcal{L}_P(\theta) \quad (1)$$

where $\mathcal{L}_{MSE}(\theta)$ represents the mean square error based on observational data (i.e., the data constraint), $\mathcal{L}_P(\theta)$ denotes the residual loss derived from the governing physical equations (i.e., the physical constraint), and $\lambda \in (0,1)$ is a hyperparameter that balances the two terms.

In this study, the physical constraint based on the Equation of State of Seawater is introduced to construct a PINN framework suitable for the joint reconstruction of oceanic temperature, salinity, and density fields. In the ocean, temperature (T), salinity (S) and density (ρ) are strongly interrelated state variables, with their dependence defined by an internationally accepted empirical relationship, the seawater equation of state: $\rho = \rho(T, S)$, where the pressure term is neglected to provide an approximate but effective physical constraint (Chen & Millero, 1977). This equation implies that any two variables uniquely determine the third, providing a robust thermodynamic prior for building a physically consistent multi-variable ocean field reconstruction model.

Crucially, the seawater equation of state exhibits mathematical properties of analytical simplicity, smoothness, and differentiability, allowing seamless integration into gradient-based deep learning optimization. Through automatic differentiation, gradients of the physical constraint can be computed efficiently and stably, avoiding numerical differentiation instability without significantly increasing computational cost.

Specifically, the seawater equation of state is formulated as:

$$\rho = \rho_w + (AS + BS^{3/2} + CS^2)$$

where

$$\begin{aligned} A &= 8.24 \times 10^{-1} - 4.08 \times 10^{-3}T + 7.64 \times 10^{-5}T^2 - 8.24 \times 10^{-7}T^3 \\ &\quad + 5.38 \times 10^{-9}T^4 \end{aligned}$$

$$B = -5.72 \times 10^{-3} + 1.02 \times 10^{-4}T - 1.65 \times 10^{-6}T^2$$

$$C = 4.83 \times 10^{-4}$$

$$\begin{aligned} \rho_w &= 999.84 + 6.79 \times 10^{-2}T - 9.09 \times 10^{-3}T^2 + 1.00 \times 10^{-4}T^3 \\ &\quad - 1.12 \times 10^{-6}T^4 + 6.53 \times 10^{-9}T^5 \end{aligned}$$

Accordingly, the physical loss term can be expressed as:

$$L_{\rho} = \sum \left| \mathbf{W}[\boldsymbol{\rho} - \boldsymbol{\rho}_w(\mathbf{A}\mathbf{S} + \mathbf{B}\mathbf{S}^{\frac{3}{2}} + C\mathbf{S}^2)] \right|^2$$

where $\mathbf{T}$, $\mathbf{S}$, $\boldsymbol{\rho}$ are all network outputs, and $\mathbf{W}$ denotes a spatially distributed weighting field.

2.2.2 Observation-Guided PINN (OG-PINN)

Previous studies have shown that the composite loss function in PINNs often exhibit magnitude disparities or conflicting optimization objectives, leading to unstable convergence (Krishnapriyan et al., 2021; Seo, 2024). In this study, since the seawater equation of state is highly nonlinear, small prediction errors in variables such as STA or SDA can be amplified in the physical loss, intensifying the competition between the data and physics terms and further impeding convergence.

To address this, we propose an Observation-Guided PINN (OG-PINN) framework. This approach introduces partial observational information into the physical loss term, aligning the optimization directions of data-driven and physics-informed components. The modified physical loss function is defined as:

$$L_{\rho} = \sum \left| \mathbf{W}[\boldsymbol{\rho}_{obs} - \boldsymbol{\rho}_w(\mathbf{A}\mathbf{S} + \mathbf{B}\mathbf{S}^{\frac{3}{2}} + C\mathbf{S}^2)] \right|^2$$

Where $\mathbf{T}$ and $\mathbf{S}$ are still predicted by the neural network, but $\boldsymbol{\rho}$ is computed using the true density values from training labels. By referencing real density in the constraint term, the network is guided to learn physically consistent T - S relationships that naturally converge toward true ocean states, thereby enhancing the model's physical consistency.

2.2.3 Adaptive Weighting Mechanism

To automatically balance the relative importance of the data-fitting and physical constraints during training, a learnable adaptive weighting mechanism is introduced. The total loss function follows Equation (1), where the physics constraint weight $\lambda = \sigma(\omega)$ is obtained by mapping a trainable scalar parameter ω through a Sigmoid activation function, ensuring $\lambda \in (0,1)$. This parameter is optimized jointly with the main U-Net in an end-to-end manner. At the early stage of training, data fitting dominates; as iteration proceeds, λ radually increases, strengthening the effect of the physics constraint and eventually balancing the magnitudes of both terms. This design eliminates laborious manual hyperparameter tuning while improving training stability and physical consistency. When the physics constraint is difficult to satisfy, the optimizer automatically reduces λ to prioritize data accuracy; conversely, when data fitting achieves low error but lacks physical consistency, λ increases adaptively to reinforce the physical regularization.

Furthermore, To better account for the spatial variability of physical processes in the ocean, we design a spatially adaptive physics constraint weight field that dynamically adjusts the influence of physics loss across geographic locations.

For a spatial domain of size $H \times W \times C$, the spatial weighting field $\mathbf{W}$ is constructed as follows:

First, Define a learnable parameter matrix $\mathbf{L} \in \mathbb{R}^{H \times W}$ and apply an exponential transform to ensure positivity:

$$\mathbf{W}_{base} = \exp(\mathbf{L})$$

which guarantees $\mathbf{W}_{base} > 0$ for all elements, maintaining gradient stability and avoiding vanishing gradients during optimization.

Next, structured geographic biases are introduced to enhance spatial pattern representation. Define a latitude bias vector $\mathbf{B}_{lat} \in \mathbb{R}^{H \times 1}$ and a longitude bias vector $\mathbf{B}_{lon} \in \mathbb{R}^{1 \times W}$. A 2D bias field is constructed as:

$$\mathbf{B} = \mathbf{B}_{lat} \mathbf{1}_{W \times 1}^T + \mathbf{B}_{lon} \mathbf{1}_{1 \times H}^T$$

Where $\mathbf{1}$ denotes a vector of ones. The bias field is scaled by $\alpha = 0.1$, to control its influence:

$$\mathbf{W}_{2d} = \mathbf{W}_{base} \odot \exp(\alpha \mathbf{B})$$

The resulting matrix $\mathbf{W}_{2d} \in \mathbb{R}^{H \times W}$ is shared across all input channels and replicated along the depth dimension to form a 3D weight tensor $\mathbf{W} \in \mathbb{R}^{H \times W \times C}$, where C is the number of channels. All parameters are initialized to zero, with $\mathbf{W}$ initialized as an all-ones matrix.

The final spatial weight map is composed of the base weights and spatial bias terms. This design not only ensures the non-negativity and spatial continuity of the weights but also significantly enhances the model's ability to represent geophysical prior structures (e.g., equatorial currents, western boundary intensification), while avoiding the oversimplification of complex ocean dynamics that typically caused by fixed or uniform weighting strategies. The result demonstrate that this mechanism effectively guides the neural network to impose stronger constraints in physically credible regions, thereby improving model generalization. The weight generation mechanism offers the following advantages: (1) Non-negative weights prevent numerical instability; (2) Separated latitude and longitude biases encode geographic prior knowledge; (3) the parameterization is concise and efficient, suitable for high-resolution grid computations.

2.3 Experimental Dataset

This study focuses on the global tropical ocean (30°S–30°N, 180°W–180°E). Model training and Validation are conducted using the Simple Ocean Data Assimilation (SODA v2.2.4) reanalysis dataset (Carton et al., 2018), which provides monthly ocean state variables including temperature (T), salinity (S), sea surface height (SSH), and vertical velocity (w) at 0.5°×0.5° spatial resolution and 14 vertical layers (0–200 m depth) for 1940–2010. The data are divided into three subsets: 80% for training (1940–1995), 10% for validation (1996–2002), and 10% for testing (2003–2009).

To ensure spatial consistency across multi-source inputs, all variables are interpolated to a $1^\circ \times 1^\circ$ grid while retaining SODA's original 14-layer vertical structure. Model training uses anomaly fields, where seasonal cycles (1940–2002 climatology) are removed to enhance sensitivity to interannual variability. Anomalies are standardized independently at each depth layer to preserve vertical structure and mitigate scale discrepancies.

Notably, the density field is not directly provided by SODA but is calculated offline using the Equation of State of Seawater – 2010 (TEOS-10) (Roquet et al., 2015) based on corresponding T, S and pressure derived from depth, ensuring strict thermodynamic consistency among the three state variables

To assess whether the physics-constrained reconstruction significantly outperforms the baseline model, we apply the correlation coefficient Z-test proposed by Meng et al. (1992). This method uses Fisher z-transformation and explicitly accounts for inter-model correlation, producing robust Z-statistic and p-value fields across the domain (Meng & Rosenthal, 1992).

3. Experimental Setup and Model Training

To systematically evaluate the performance of the proposed OG-PINN framework in reconstructing oceanic temperature and salinity fields, three comparative experiments were designed:

- (1) a multi-task U-Net baseline model (Benchmark, without physical constraints);
- (2) a standard PINN, which incorporates the seawater equation of state as a physical constraint based on the same baseline architecture;
- (3) the proposed Observation-Guided PINN (OG-PINN)

All three experiments adopt the CBAM-attention multi-task U-Net as the base architecture to ensure structural comparability. Detailed configurations are provided in Table 1.

Table 1 Configuration details of model parameters.

	Loss function weighth	Optimizer	Activation	Learning rate	Training samples	Epoch
UNet	——					
PINN	Adaptive weights	Adam	ReLU	ReduceLROnPlateau	1940-2002	300
OG-PINN	Adaptive weights					

By analyzing the evolution of loss functions during training (Fig. 2), we reveal the competitive relationship between data and physical constraints in the PINN framework. Figure 2 show that the trends of the total loss, data constraint loss, and physics constraint loss on both training and validation sets for PINN and OG-PINN.

In the standard PINN (Fig. 2b), the data loss converges around epoch 100 and stabilizes around 1.0. However, the physics loss (unweighted raw residual loss from the Equation of State of Seawater) exhibits significant instability during training (Fig. 2c)

and even upward trends on the training set. This indicates that the model fails to learn the thermodynamic laws governed by the seawater equation of state.

In contrast, the OG-PINN demonstrates a more stable training behavior. Its data loss similarly converges around 1.0 after about 100 epochs (Fig. 2e), while its physics loss steadily decreases and stabilizes at ~ 0.04 (Fig. 2f), indicating consistent convergence of both components. Notably, although the final physics loss in standard PINN appears numerically smaller (below 0.04), this represents false convergence: the outputs formally yield small residuals but fail to capture physically consistent T–S relationship. The reconstructed fields deviate from realistic ocean states, degrading reconstruction accuracy.

By introducing observational guidance, OG-PINN effectively harmonizes the optimization directions of data and physics constraints, achieving coherent convergence during training. This result clearly verifies the superiority of OG-PINN in improving both the physical plausibility and reconstruction accuracy.

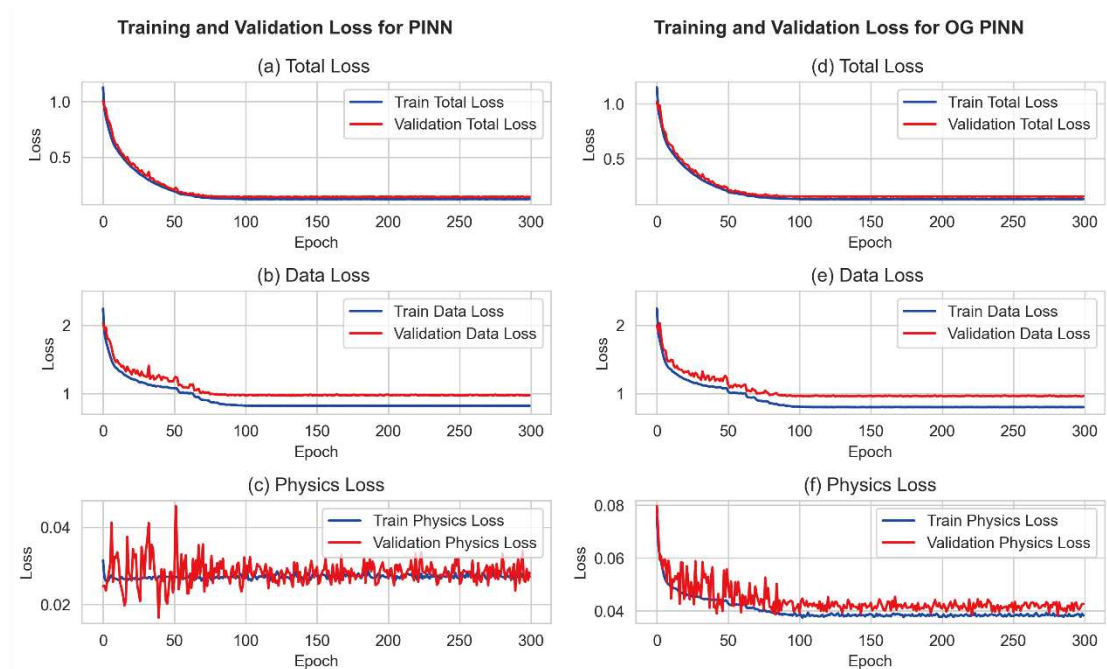


Figure 2 Evolution of loss functions during model training. (a–c) represent the total loss, data loss, and unweighted original physical loss of the PINN, respectively. (d–f) show the corresponding loss functions of the OG-PINN.

4. Comparison of Reconstruction Results

4.1 Reconstruction Performance

Figure 3 presents the spatial distribution of correlation coefficients between reconstructed subsurface temperature anomalies (STA) and corresponding reanalysis data at depths of 95 m and 195 m for the three networks. At 95 m depth (Fig. 3a–c), all models (OG-PINN, PINN, and U-Net) exhibit high correlations, particularly in the

tropical Pacific and Indian Ocean, where coefficients generally exceed 0.7. Over the entire tropical region (30°S–30°N), the mean correlation coefficients are 0.733 (OG-PINN), 0.731 (PINN), and 0.725 (U-Net), indicating robust STA reconstruction performance at this depth. At 195 m (Fig. 3d–f), the overall correlation weakens but remains above 0.6 in most regions, particularly in the off-equatorial Pacific and the western Pacific warm pool. The tropical mean correlations drop to 0.610 (OG-PINN), 0.606 (PINN), and 0.599 (U-Net).

Notably, low-correlation areas become more pronounced as depth increases, particularly in the tropical Atlantic, the eastern Pacific cold tongue, and the upwelling zone of the eastern Indian Ocean. In the tropical Atlantic, the Atlantic Niño signal is relatively weak, and the Bjerknes feedback is more unstable than its Pacific counterpart (Chen et al., 2024; Dippe et al., 2019). Thus, subsurface temperature variability is considerably modulated by upwelling and mesoscale dynamics, resulting in lower STA reconstruction accuracy than in the Pacific. In the eastern Pacific and eastern Indian Ocean, intense equatorial and coastal upwelling dominate the vertical structure, making it difficult to fully capture the three-dimensional thermohaline characteristics using only surface information (Chiang et al., 2022; Y. Li et al., 2017).

Overall, all three models effectively capture large-scale STA structures in the tropical ocean, validating the feasibility of reconstructing subsurface thermodynamic states from surface data.

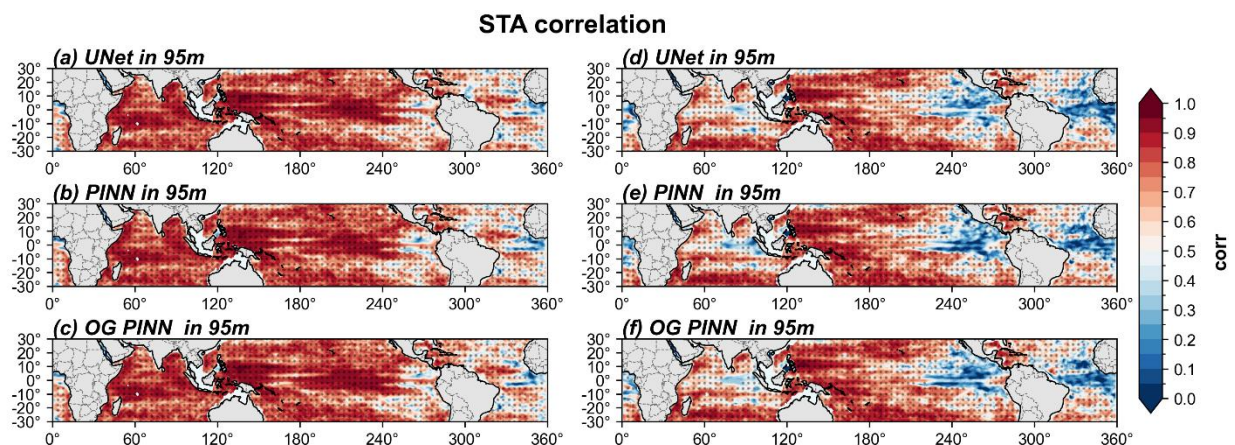


Figure 3 Spatial distribution of correlation coefficients between reconstructed and SODA dataset subsurface temperature anomalies (STA) at depths of 95 m (a – c) and 195 m (d – f) for the testing dataset, as produced by U-Net, PINN, and OG-PINN. Dotted areas indicate regions where correlations pass the significance test ($p < 0.05$).

Further analyzing the vertical distribution of root mean square error (RMSE), Figure 4 shows the global tropical average and basin-specific vertical RMSE profiles of STA (Fig. 4a–d), as well as the differences between U-Net and PINN/OG-PINN (Fig. 4e–h). OG-PINN achieves the lowest RMSE values in all basins. On the global scale, the maximum RMSEs are 0.49 °C (OG-PINN), 0.50 °C (PINN), and 0.51 °C (U-Net), all peaking near 115 m, the depth corresponding to the dynamically active thermocline

(Fig. 4a). At the tropical, both PINN and OG-PINN effectively reduce STA reconstruction RMSE at all depths. Notably, OG-PINN shows the most significant RMSE reduction in the 50–120 m range (Fig. 4e), indicating that physical constraints enable accurate representation of thermocline dynamics. In contrast, PINN exhibits less depth-dependent RMSE reduction (Fig. 4e), reflecting its insufficient characterization of physical processes near the thermocline.

The Indian Ocean STA RMSE is not only the lowest in absolute value but also shows the smallest inter-model performance differences, with error peaks at 75 m (Fig. 4d). In contrast, the Pacific and Atlantic exhibit RMSE peaks at 125 m (Fig. 4b,c). This contrast highlights fundamental basin-scale dynamical differences: the Indian Ocean’s shallow thermocline, dominated by monsoon forcing, exhibits low STA variability but complex dynamics. These dynamics limit OG-PINN’s improvement over U-Net to less than 0.01°C at all depths (Fig. 4h).

By comparison, the Pacific and Atlantic Oceans exhibit greater thermocline depths and are accompanied by stronger oceanic dynamic signals (e.g., internal waves, Rossby waves, and interannual variability), with their thermal structures governed by oceanic internal dynamic processes. Here, statistical models is insufficient to resolve fine thermocline structures, while physics-informed networks (especially OG-PINN) significantly reduce STA RMSE by enforcing thermodynamic consistency. In the Pacific, OG-PINN reduces RMSE by over 0.02°C at 50–120 m; in the Atlantic, RMSE decreases by nearly 0.03°C below 100 m. PINN also reduces RMSE in most regions but less effectively than OG-PINN.

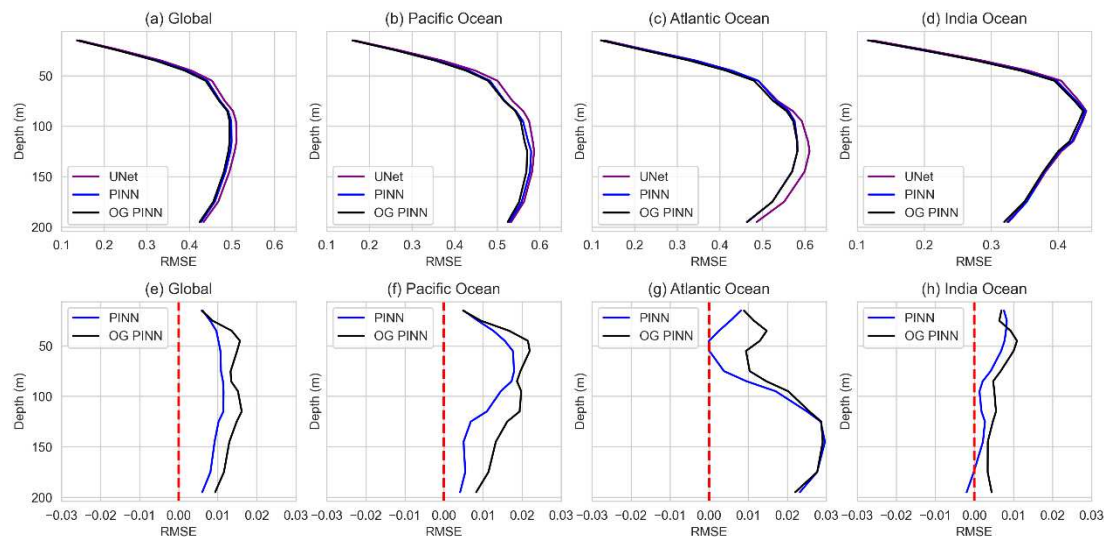


Figure 4 (a–d) present the vertical profiles of the regional mean root-mean-square error (RMSE) of STA across the global tropics and three major ocean basins (tropical Pacific, Indian, and Atlantic Oceans) for U-Net, PINN, and OG-PINN in the testing dataset. (e–h) is the vertical distribution of RMSE differences between PINN/OG-PINN and U-Net, calculated as U-Net minus PINN/OG-PINN. Positive values indicate superior performance of PINN/OG-PINN over U-Net, while negative values suggest the opposite.

However, the SSA exhibits distinct characteristics. Unlike temperature, SSA distributions are governed by evaporation-precipitation cycles, ocean circulation, freshwater input, and mixing processes, resulting in weaker and more nonlinear coupling with surface states (Köhler et al., 2018; Yu, 2014; Zhang et al., 2023). Figure 5 shows the spatial correlation between reconstructed SSA and reanalysis data at 95 m and 195 m. Compared to STA (Fig. 5a–c), SSA correlations are significantly lower: at 95 m, the mean coefficients are 0.486 (OG-PINN), 0.478 (PINN), and 0.474 (U-Net), respectively, dropping to 0.19–0.20 at 195 m (Fig. 5d–f). Above 75 m, SSA correlations often exceed 0.5, but with increasing depth, high-correlation zones are confined to mid-latitude regions (10° S–25° S). The equatorial band (10° S–10° N), heavily affected by precipitation and complex dynamics, remains hardest to reconstruct.

This systematic underperformance in salinity reconstruction is not unique to this study but a common challenge in ocean data reconstruction. Independent studies all report this challenge: Song et al. (2024) observed rapid SSA RMSE increases below 50 m in the tropical Pacific; Dong et al. (2024) found SSA correlations in the South China Sea decayed sharply below 50 m, reaching minima at 70 m; Meng et al. (2024) highlighted SSA’s lower reconstruction accuracy compared to temperature in the Pacific. These results collectively indicate that salinity’s multiscale hydrological and dynamical controls inherently limit its reconstructability.

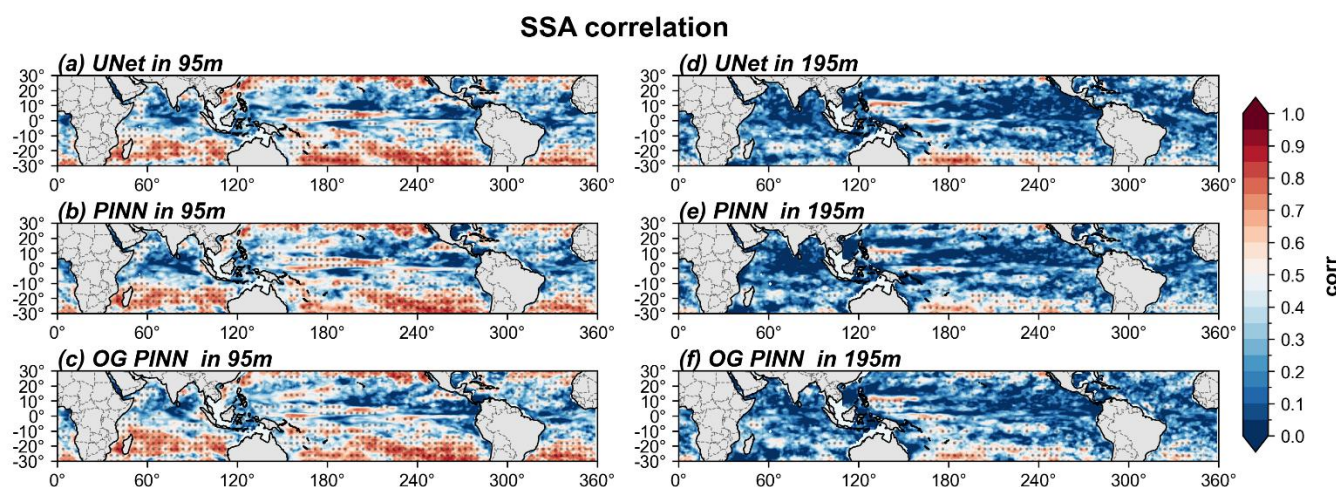


Figure 5 (a–c) and (d–f) show the spatial distribution of correlation coefficients between reconstructed and SODA dataset subsurface salinity anomalies (SSA) at depths of 95 m and 195 m, respectively. Dotted areas indicate regions where correlations pass the significance test ($p < 0.05$).

To quantify SSA performance across depths and basins, Figure 6 presents the vertical RMSE distributions for tropical and basin average SSA reconstructions (Fig. 6a–d) and improvements over U-Net (Fig. 6e–h). Globally, OG-PINN achieves the lowest RMSEs, with a maximum of 0.108 psu, compared to 0.111 psu (PINN) and 0.109 psu (U-Net). All models exhibit their RMSE peaks around 115 m, near the thermocline (Fig. 6a). This indicates that, although SSA variability is low, the steep vertical gradients near the thermocline, induced by multiscale processes such as mixing,

advection, and precipitation-evaporation (P-E), increase reconstruction difficulty. On the tropics, PINN reduces RMSE only above 50 m, whereas OG-PINN shows significant improvements above 120 m and outperforms PINN. Below 120 m, OG-PINN's performance becomes comparable to U-Net (Fig. 6e). Unlike temperature, SSA RMSE improvements near the thermocline are less pronounced, reflecting its higher dynamical and thermodynamic complexity.

At basin-scale the Indian Ocean's SSA RMSE peak occurs at 85 m (Fig. 6d), while the Pacific and Atlantic exhibit deeper peaks at 125 m (Fig. 6b,c). Except below 125 m in the Pacific and 200 m in the Indian Ocean, OG-PINN consistently outperforms U-Net, with RMSE reductions exceeding 0.005 psu. In contrast, PINN performs well only in the Atlantic.

Overall, OG-PINN demonstrates superior global SSA reconstruction but its physical constraint effectiveness diminishes in deeper layers, reflecting SSA's higher dynamical complexity. The importance of refined coordination between physical and data constraints in these regions.

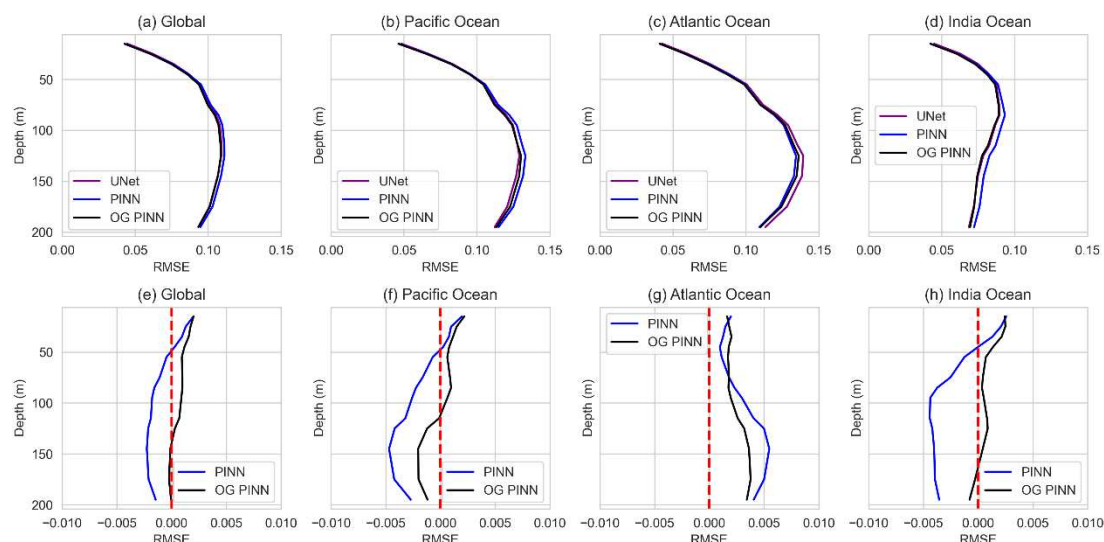


Figure 6 (a–d) is the vertical profiles of regional mean RMSE for SSA across the global tropics and three major ocean basins (tropical Pacific, Indian, and Atlantic Oceans) for U-Net, PINN, and OG-PINN in the testing dataset. (e–h) is the vertical distribution of RMSE differences between PINN/OG-PINN and U-Net.

Notably, since ocean density in the tropical upper ocean (0–200 m) is primarily temperature-driven (thermal expansion dominates over salinity), the spatial correlation patterns of reconstructed SDA similar to STA results. Correlation coefficients generally exceed 0.7 above 100 m and 0.6 at 200 m, with the highest values in the eastern Pacific. Correlations decline sharply in the tropical Atlantic, eastern Pacific cold tongue, and eastern Indian Ocean upwelling zones with increasing depth (not shown).

However, the vertical RMSE profiles of SDA reconstruction differ from STA (Fig. 7). Across the tropics, all models exhibit maximum RMSE at 75 m significantly

shallower than STA's peak (115–125 m), with values of 0.145 kg/m³ (OG-PINN), 0.146 kg/m³ (PINN), and 0.149 kg/m³ (U-Net) (Fig. 7a). OG-PINN's most significant improvements occur around 50 m, while below 100 m, OG-PINN and PINN RMSEs approach U-Net levels (Fig. 7e). This pattern is consistent across basins: OG-PINN shows peak improvements at around 50 m, with the Pacific exhibiting the largest RMSE reduction (0.008 kg/m³). Standard PINN underperforms OG-PINN in most cases, and in the upper Atlantic (<100 m), its physical constraint even increases RMSE. Only below 100 m in the Atlantic does PINN slightly outperform OG-PINN.

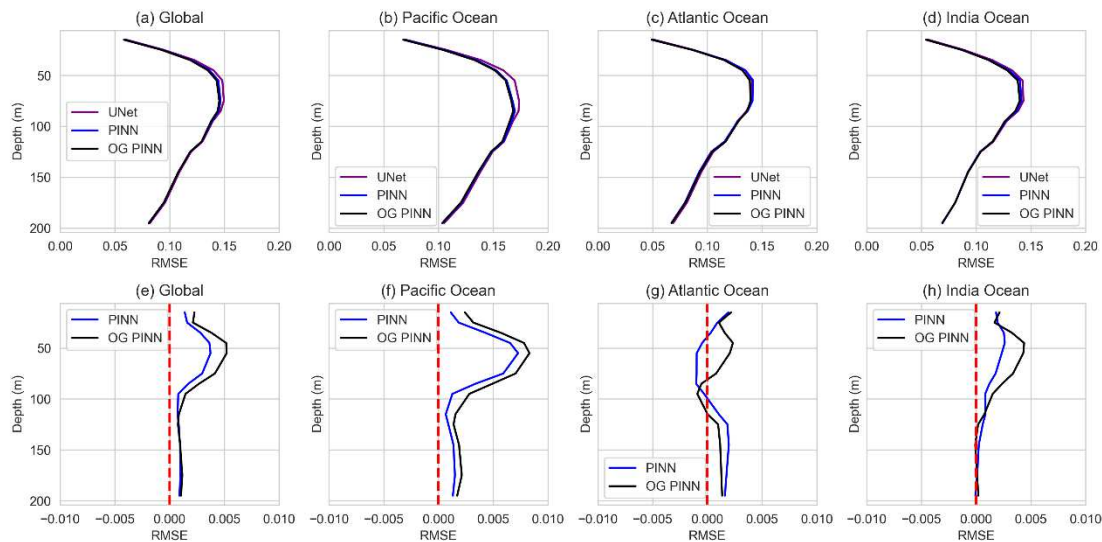


Figure 7 (a–d) is the vertical profiles of regional mean RMSE for subsurface density anomalies (SDA) across the global tropics and three major ocean basins (tropical Pacific, Indian, and Atlantic Oceans) for U-Net, PINN, and OG-PINN in the testing dataset. (e–h) is the vertical distribution of RMSE differences between PINN/OG-PINN and U-Net.

In summary, this section systematically evaluates subsurface reconstruction performance for temperature, salinity, and density. STA, strongly coupled to surface signals, achieves high accuracy (correlation > 0.6) at 100–200 m, with OG-PINN significantly outperforming benchmarks near the thermocline (~115 m), particularly in the Pacific and Atlantic. SSA exhibits lower reconstruction skill, with correlations dropping to ~0.2 at 200 m and physical constraints effective only above 120 m, reflecting its more complex dynamical control. SDA, dominated by STA, shares similar spatial correlation patterns but peaks at shallower RMSE depths (75 m). Notably, standard PINN often performs unstably in SSA reconstruction due to conflicts between physical and data constraints, particularly in the Atlantic and Indian Ocean, while OG-PINN's observational guidance ensures robust improvements. These results demonstrate that ocean variable reconstruction skill depends on their governing physical mechanisms, and OG-PINN, by adaptively balancing physical and data constraints, achieves superior and more stable reconstruction performance.

4.2 Spatial Patterns of Performance Improvement in OG-PINN

To quantify the spatial characteristics of OG-PINN improvements in STA reconstruction, Figure 8 displays the spatial distribution of correlation-coefficient differences between OG-PINN and U-Net (Figure 8a,d), as well as between OG-PINN and standard PINN (Figure 8b,e), at depths of 95 m and 195 m (only regions with statistically significant differences are shown).

At 95 m depth, OG-PINN significantly outperforms U-Net across most ocean basins, particularly in the Niño3.4 region and the subtropical Atlantic, where the maximum correlation improvement reaches 0.42. Compared to standard PINN, OG-PINN also demonstrates notable advantages, especially in the central Pacific (maximum correlation improvement of 0.30).

It is noteworthy that both physics-constrained models (OG-PINN and PINN) exhibit performance degradation in strong upwelling regions, including the eastern Pacific cold tongue (5°S – 5°N , 100°W – 120°W), the equatorial eastern Atlantic (0° – 10°N , 20°W – 0°), and the equatorial Indian Ocean (5°S – 5°N) (Figure 8a,b). However, OG-PINN still performs significantly better than standard PINN in these regions, particularly in the cold tongue area, indicating that its observation-guided mechanism effectively mitigates physical constraint inaccuracies (Figure 8c). Additionally, in non-upwelling regions such as the central Pacific, the northern coast of South America, and the south of the Indian Peninsula, OG-PINN further enhances reconstruction accuracy, though these improvements are not statistically significant.

The spatial patterns of performance degradation in PINN and OG-PINN relative to U-Net (Figure 8a,b) align with the climatological mean distribution of vertical velocity (w , Figure 9a). The regions with the most severe performance degradation correspond precisely to areas of the strongest climatological upwelling. This suggests that in these regions, the subsurface thermodynamic structure is primarily governed by localized vertical transport (e.g., Ekman suction, thermocline shoaling), which weakens the correlation with surface states and introduces highly nonlinear temperature-salinity (T-S) relationships. These factors pose significant challenges to model physical consistency and generalization capability.

At 195 m depth, the advantage of OG-PINN over U-Net becomes more pronounced, particularly in the central tropical Pacific and the off-equatorial Atlantic, where the maximum correlation improvement reaches 0.55 (Figure 8d). This highlights OG-PINN's enhanced capacity to capture deep thermodynamic structures. However, performance degradation is also more severe in the core upwelling zones (cold tongue, equatorial eastern Atlantic Ocean, equatorial Indian Ocean). In contrast, standard PINN, while slightly outperforming U-Net in most regions, exhibits more severe performance deterioration in upwelling zones (Figure 8e).

Critically, the performance improvement of OG-PINN over PINN are spatially concentrated in the core upwelling regions, with a maximum correlation improvement of 0.58, coinciding with regions of large negative vertical velocity w (downwelling defined as positive) (Figure 8f). This directly validates that by incorporating observational data into physical constraints, OG-PINN better reflects the true ocean

state and enhances the model's responsiveness to key dynamical processes such as upwelling. Consequently, OG-PINN achieves superior physical consistency and reconstruction accuracy in regions dominated by complex dynamics.

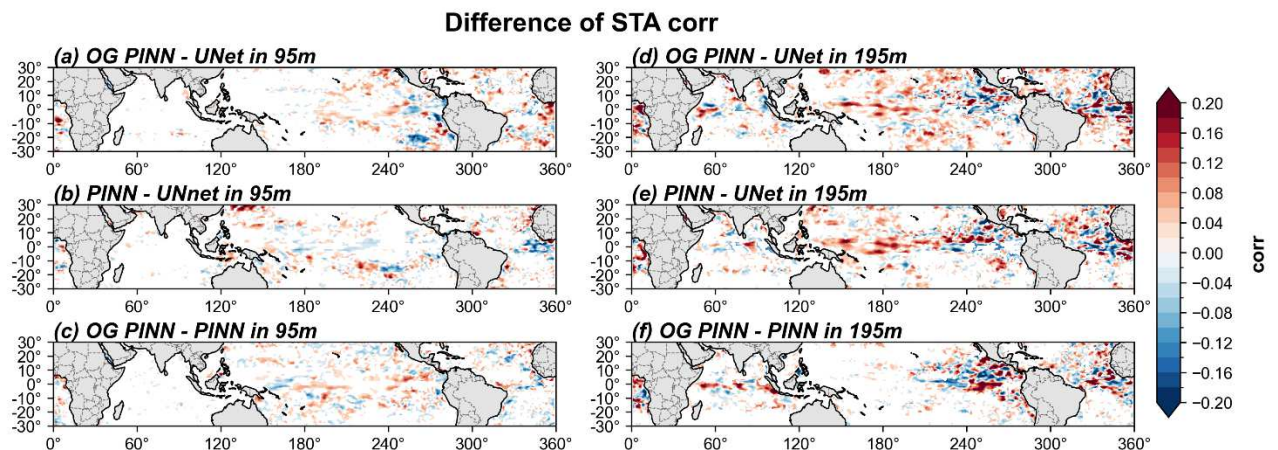


Figure 8 Spatial distribution of correlation coefficient differences between reconstructed and SODA STA fields at 95 m (a–c) and 195 m (d–f). (a, d) OG-PINN minus U-Net; (b, e) PINN minus U-Net; (c, f) OG-PINN minus PINN. Only regions passing the statistical significance test are displayed.

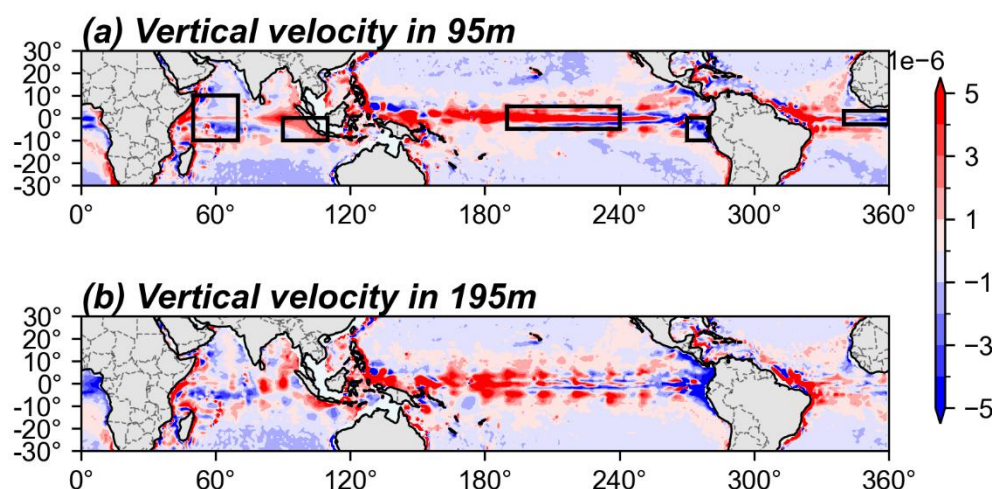


Figure 9 Climatological distribution of vertical velocity (w) from SODA data at 95 m (a) and 195 m (b), with downward motion defined as positive.

In (a), the black boxes from left to right denote the regions corresponding to EIO, WIO, Niño 3.4, Niño 1+2, and ATL3.

To further evaluate model performance in key climate mode regions, we selected five representative climate indices: strong upwelling regions, Niño1+2 (eastern Pacific cold tongue), the western pole of the Indian Ocean Dipole (WIO, western Indian Ocean); and weak/non-upwelling regions — Niño3.4 (central Pacific), ATL3 (equatorial eastern Atlantic), and the eastern pole of the Indian Ocean Dipole (EIO, eastern Indian Ocean).

The specific locations are shown in Figure 9a. We calculated the correlation between reconstructed and ground truth STA at varying depths in these regions (Figure 10).

The results demonstrate that in Niño3.4 and ATL3, OG-PINN significantly outperforms other models across the entire water column (15–195 m). In particular, performance improvements in the Atlantic ATL3 region are most pronounced, indicating that physics constraints helps more accurately describe the complex subsurface thermal structure in this dynamically active Atlantic zone. However, in EIO, which lacks persistent upwelling but is influenced by monsoon-driven dynamics, OG-PINN performs comparably to U-Net, with slight degradation at deeper levels.

Further analysis reveals that in strong upwelling regions (Niño1+2, WIO), OG-PINN maintains superiority only in the upper layer (shallow than 50 m). As depth increases (>50 m), its performance rapidly declines, even underperforming the purely data-driven U-Net. This phenomenon highlights the fundamental cause of spatial variability in OG-PINN’s performance improvement: deep water masses originate from cold sources below the thermocline, governed by dynamical processes such as vertical mixing and Ekman suction, which decouple from surface signals. In such regions, despite incorporating seawater equation of state constraints, the physics loss term fails to reflect upwelling dynamics and may even drive the model toward unphysical solutions, leading to performance degradation.

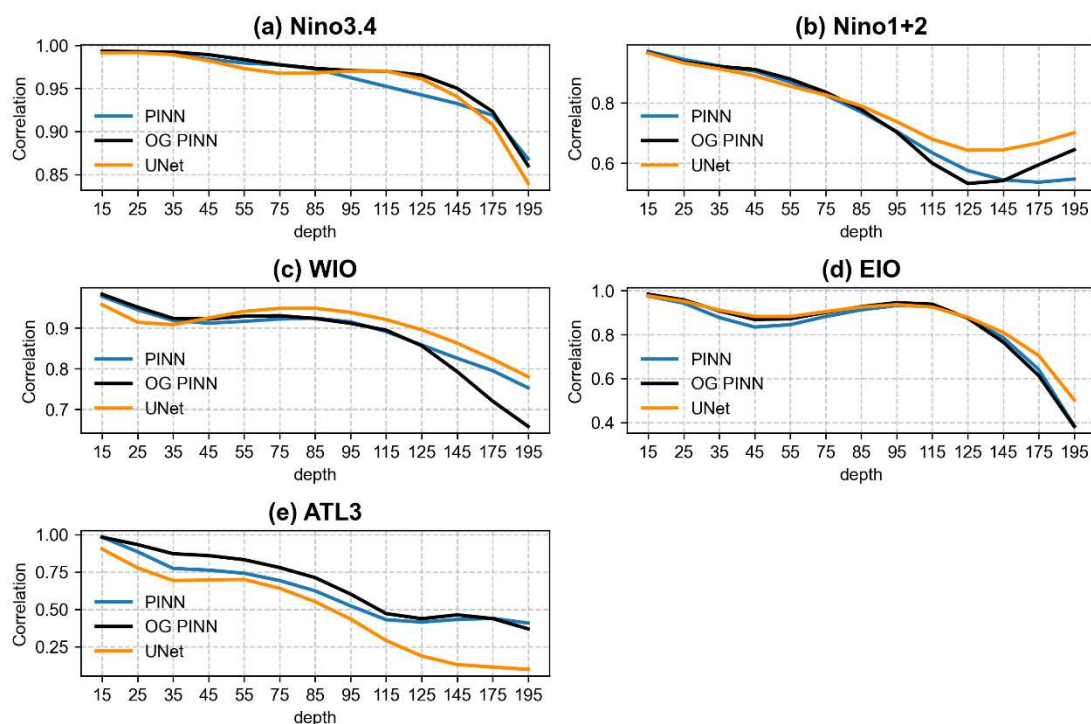


Figure 10 Correlation between reconstructed and SODA climate indices across depths. (a) Niño 3.4, (b) Niño 1+2, (c) WIO, (d) EIO, (e) ATL3.

Compared with STA, SSA is jointly regulated by evaporation-precipitation, freshwater input, and oceanic mixing processes, with stronger coupling between spatial variability and dynamical mechanisms. As depth increases, SSA reconstruction accuracy deteriorates significantly. Therefore, comparisons of model performance must

focus on depth layers where correlations pass statistical significance tests. Unlike STA analysis, SSA reconstruction is evaluated at 55 m and 95 m depth levels.

Figure 11 shows the spatial distribution of correlation-coefficient differences between reconstructed and ground truth SSA at 95 m and 195 m depths. Results demonstrate that OG-PINN exhibits significant advantages in SSA reconstruction. As shown in Figure 11a–c, at 55 m depth, OG-PINN improves the correlation with reanalysis in the warm pool region, equatorial western Pacific, and north of the equatorial Atlantic compared to U-Net. Notably, correlations improve markedly north of Australia, with a maximum increase of 0.36 (Figure 11a). However, performance slightly degrades near the Philippine Islands, east of Australia (East Australia Current region), the Pacific cold tongue, equatorial Indian Ocean, and equatorial Atlantic (Figure 11a). Standard PINN shows similar spatial improvement patterns to OG-PINN relative to U-Net, concentrated in the northeastern Australian waters and north of the equatorial Atlantic. Additionally, PINN improves SSA reconstruction off the Chilean coast. Nevertheless, significant degradation occurs east of Australia in PINN (Figure 11b). A direct comparison between OG-PINN and PINN reveals that the former outperforms the latter in extratropical regions, particularly east of Australia, where correlations improve by up to 0.38 (Figure 11c).

At 95 m depth, OG-PINN shows a pattern similar to that at 55 m: correlations improve significantly in the warm pool region, with a maximum increase of 0.55, but degrade in the cold tongue upwelling zone, western boundary current regions of the Pacific, and east of Australia (Figure 11d). In contrast, PINN exhibits additional correlation degradation in parts of the Indian Ocean (Figure 11e). OG-PINN's improvements over PINN are primarily concentrated in the southern Indian Ocean, extratropical Pacific, and extratropical Atlantic, with maximum correlation increase reaches 0.65 (Figure 11f).

Notably, unlike STA reconstruction, both PINN methods show significant performance degradation not only in upwelling regions but also in Pacific western boundary current zones (e.g., Kuroshio and East Australia Current). This phenomenon may stem from complex dynamical processes in these region, intense vertical exchange, frontal activity, and nonlinear advection, which make it difficult for the seawater equation of state to capture realistic physical constraints, potentially introducing additional errors instead.

SDA reconstruction performance closely resembles STA: both PINN variants degrade significantly in upwelling regions but generally improve correlations relative to U-Net elsewhere. However, due to OG-PINN's physical constraints not directly acting on the neural network's output density field (as shown in Equation 5), its improvements for SDA are notably smaller than for STA, and even slightly inferior to standard PINN in parts of the Atlantic (figures not shown). Given the high spatial overlap with STA and space constraints, this is briefly mentioned here.

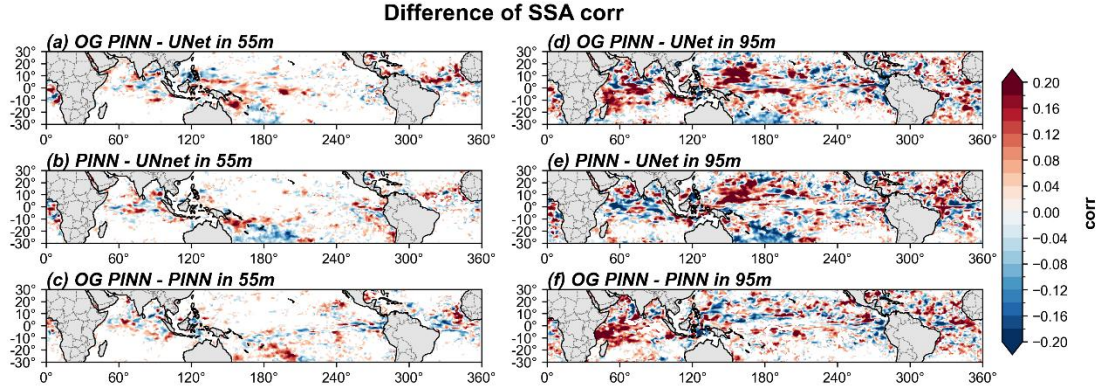


Figure 11 Spatial distribution of correlation coefficient differences between reconstructed and SODA SSA fields at 95 m (a–c) and 195 m (d–f). (a, d) OG-PINN minus U-Net; (b, e) PINN minus U-Net; (c, f) OG-PINN minus PINN. Only regions passing the statistical significance test are displayed.

To more precisely characterize the spatial variability in the effectiveness of physical constraints, we introduced a spatially adaptive weighting mechanism in both PINN and OG-PINN. This mechanism allows each model to dynamically adapt the weight of the physical loss term based on local ocean dynamical features (Figure 12). It should be noted that both models adopt identical initial spatial weight parameters, but their trained weight distributions exhibit strikingly different spatial patterns: OG-PINN develops highly heterogeneous spatial structures, whereas standard PINN weights converge toward global uniformity. This divergence reveals OG-PINN’s critical role in discerning the applicability of physical constraints.

The spatial distribution of physics constraint weights demonstrates that OG-PINN can implement fine regulation of constraint strength in response to spatial heterogeneity in oceanic dynamics (Figure 12a). In equatorial upwelling zones (Pacific cold tongue, equatorial Indian Ocean, and equatorial Atlantic Ocean), the physical constraint weight generally drops below 0.83, coinciding closely with the areas where both PINN and OG-PINN exhibit degraded STA reconstruction performance (Figure 8a,b,d,e). This phenomenon has clear physical implications: intense upwelling-driven vertical transport, combined with frontal activity and nonlinear advection processes (Chelton et al., 2011; Cui et al., 2024), makes it difficult for an idealized seawater equation of state to accurately represent local thermodynamic relationships. Imposing fixed-strength constraints in these regions would introduce systematic biases, leading to reconstruction distortions. For SSA reconstruction, the sensitivity to physical processes extends to broader areas. Beyond equatorial upwelling zones, western boundary current regions (e.g., Kuroshio and East Australia Current) also exhibit marked reconstruction accuracy degradation (Figure 11a,b,d,e).

To address this, OG-PINN actively reduces constraint weights in these zones. The spatial correspondence between OG-PINN’s weight distribution and regions of performance degradation indicates that the model does not passively suppress physical constraints. Instead, it actively identifies areas where physical priors lose validity and adaptively reduces their weights to mitigate errors caused by idealized assumptions.

This spatially distributed weighting strategy clearly reflects OG-PINN's perceptual capability for regional oceanic dynamics and confirming the critical role of its observation guided mechanism under complex oceanic conditions.

In contrast, standard PINN shows an almost uniform tropical weight distribution (Figure 12b), with only slight reductions in the western Pacific and equatorial Atlantic. Although the model incorporates spatial adaptability, it tends to fall into 'spurious convergence', where the numerical physical residual decreases, while the predicted solution diverges from actual states. Consequently, it fails to identify regions where physical constraints have become invalid, unable to suppress weights effectively in upwelling and western boundary current zones. This forces the model to impose idealized constraints in complex dynamical regions, exacerbating reconstruction errors.

This result aligns closely with OG-PINN's superior performance over standard PINN in previous comparisons (Figure 8c,e; Figure 11c,e). In these dynamically complex zones, OG-PINN's reconstruction accuracy significantly outperforms standard PINN, precisely due to its spatially responsive weighting of the physical constraint. OG-PINN's adaptive weight reduction represents a rational adjustment based on observation-guided perception of real ocean states, whereas standard PINN's weight distribution fails to reflect spatial complexity in oceanic dynamics. Its physical constraint loses guiding significance due to false convergence, rendering its "adaptive" mechanism ineffective. This further confirms that effective physics-informed modeling requires not only incorporating physical laws but also their consistency with real system states.

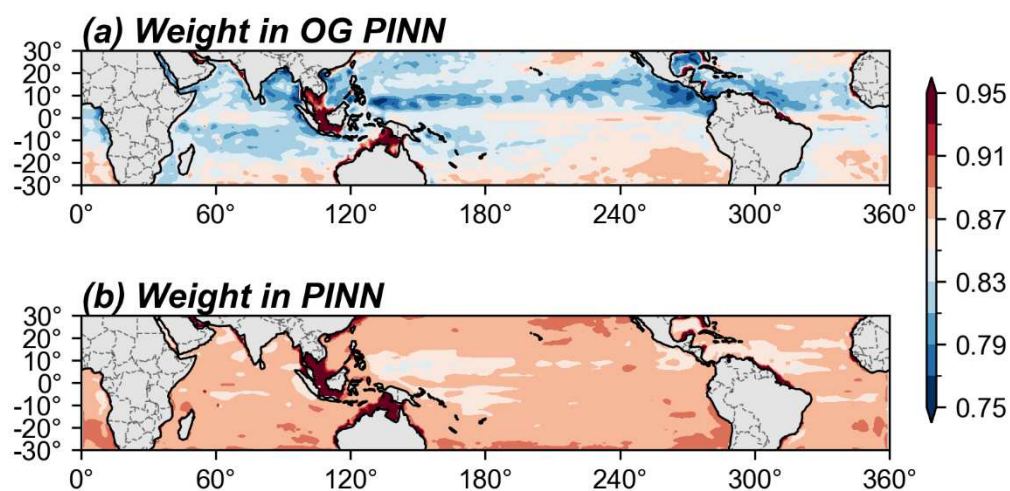


Figure 12 Spatial maps of the dynamic physics loss weight W for (a) OG-PINN and (b) PINN.

In summary, OG-PINN's performance improvements exhibit distinct spatial variability. For STA, its advantages are concentrated in the tropical central Pacific and subtropical Atlantic, with maximum correlation gains of 0.42 at 95 m depth and 0.55 at 195 m depth. For SSA, OG-PINN significantly outperforms baseline models in the warm pool region, extratropical Pacific, and Atlantic. However, in strong upwelling

zones, both physics constrained models show performance degradation in reconstructing STA and SSA.

This spatial pattern closely corresponds to the climatological vertical velocity distribution. Stronger upwelling leads to greater decoupling between subsurface and surface states. In such areas, the static equation of state fails to represent the real dynamic processes, causing the physical constraint to become invalid or even introduce bias. Standard PINN, lacking observation guidance, tends to fall into 'spurious convergence' in complex regions—satisfying equations while diverging from true states. In contrast, OG-PINN's observation-guided mechanism enhances physical consistency across most oceans and actively reduces constraint strength in dynamically sensitive zones, balancing accuracy and physical plausibility.

4.3 Dynamical Signals in the Equatorial Region

The equatorial region is one of the most dynamic and climate-sensitive zones in the global climate system, characterized by highly nonlinear ocean–atmosphere interactions and tightly coupled multi-scale processes. This region not only receives the strongest solar insolation and sustains the most vigorous convection but also serves as the primary pathway for the generation, propagation, and interaction of various oceanic–atmospheric waves, such as Kelvin waves, Rossby waves, and mixed Rossby–gravity waves (Baquero-Bernal & Latif, 2005a; Boulanger et al., 2004; Fang & Xie, 2020; Gill, 1980). These equatorial wave signals significantly influence the evolution of major climate modes, including El Niño–Southern Oscillation (ENSO), the Indian Ocean Dipole (IOD), and Atlantic Niño (Chowdary, 2013; Capotondi, 2021; Muhammad et al., 2019; Zhou & Zhang, 2023, 2025).

In this context, the T-S structure of the upper equatorial ocean (0–200 m) not only mediates air–sea exchange but also critically governs wave propagation speeds, thermocline depth variability, and subsurface heat storage (Baquero-Bernal & Latif, 2005b; Chang et al., 2003; Gill, 1980; McPhaden, 2003; Jing et al., 2025). Accurate reconstruction of equatorial temperature and salinity fields is essential for improving the predictive skill of climate models.

This section focuses on the equatorial band (5°S–5°N) and systematically evaluates the reconstruction performance of U-Net, PINN, and OG-PINN for STA, SSA and SDA. Special attention is given to their behavior in core upwelling regions and along key wave propagation pathways, to assess whether OG-PINN effectively enhances the model's ability to represent complex dynamical processes.

Figure 13c shows the meridional section of the standard deviation (STD) of observed STA across the equatorial band, revealing the spatial structure of the thermocline. In the western Pacific warm pool (100°E–180°), the thermocline is deep and stable. In contrast, in the eastern Pacific (120°W–90°W), eastern Atlantic (20°W–0°), and western Indian Ocean (40°E–70°E), persistent upwelling brings cold subsurface water to the surface, resulting in a shallower and more variable thermocline.

In the standard PINN (Figure 13a), despite the incorporation of physical constraints, the RMSE of STA reconstruction increases relative to U-Net in most

thermocline regions, particularly below 100 m in the eastern Pacific and Indian Ocean. This indicates that standard PINN fails to accurately capture the true temperature structure in this high-gradient, strongly nonlinear zone. Its physical constraint does not positively contribute to STA reconstruction but, instead, drives the optimization toward strict adherence to the seawater equation of state, thereby diverting the model solution from the true ocean state and reducing its adaptability to complex dynamical processes.

By contrast, OG-PINN (Figure 13b) achieves lower RMSE than U-Net throughout the thermocline regions of the eastern Pacific, eastern Atlantic, and Indian Ocean, demonstrating that its physics constraints better align with real dynamical processes and enhance generalization in these critical areas. More importantly, the RMSE reduction of OG-PINN relative to standard PINN (Figure 13d) shows consistent improvement across nearly the entire thermocline, with particularly pronounced improvements in the upwelling-active zones of the eastern Pacific and western Indian Ocean.

This advantage originates from OG-PINN's observation-guided physical constraint design, in which the true ocean state is embedded into the formulation as a prior, ensuring that the optimization of the physical-loss function proceeds consistently with real oceanic states. In dynamically active regions, the standard PINN tends to suffer from false convergence, wherein the physical residual becomes small but the resulting solution does not correspond to physical reality. In contrast, OG-PINN mitigates the competition between data and physics constraints. As a result, it preserves physical plausibility while simultaneously enhancing its ability to capture realistic nonlinear ocean structures.

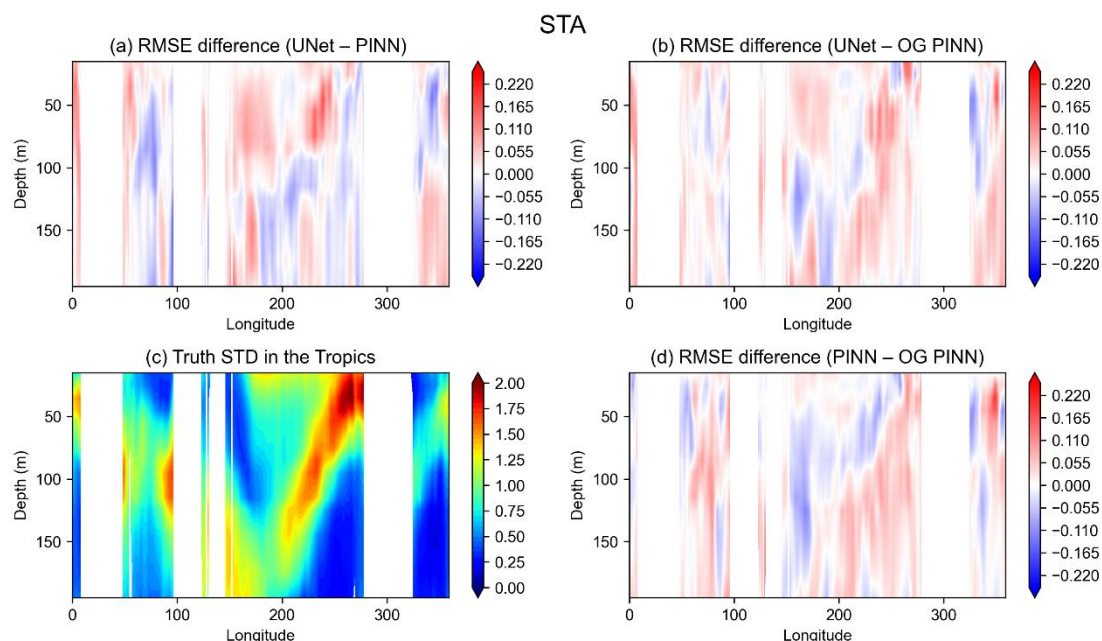


Figure 13 Comparison of STA reconstruction RMSE among different models in the equatorial band. (a) UNet minus standard PINN; (b) UNet minus OG-PINN; (c) SODA STA standard deviation (STD) over the equator, indicating the spatial pattern of natural variability; (d) PINN minus OG-PINN.

All subplots are longitude–depth sections (5°S–5°N average).

The regions with the strongest SSA variability are concentrated in the western Pacific warm pool (Figure 14c). In this area, both OG-PINN and PINN significantly reduce the root-mean-square error (RMSE), indicating that physical constraints effectively improve SSA reconstruction accuracy in high variability zones (Figures 14a, b). However, in the equatorial cold tongue and equatorial Atlantic, where intense upwelling enhances vertical mixing, the SSA structure becomes highly nonlinear. In these regions, both PINN-type models exhibit markedly degraded performance, with RMSE increasing relative to baseline methods. Furthermore, the performance improvement of OG-PINN over standard PINN is primarily confined to the western Pacific warm pool and depths below 100 m in the Indian Ocean; elsewhere, the differences between the two models are relatively minor.

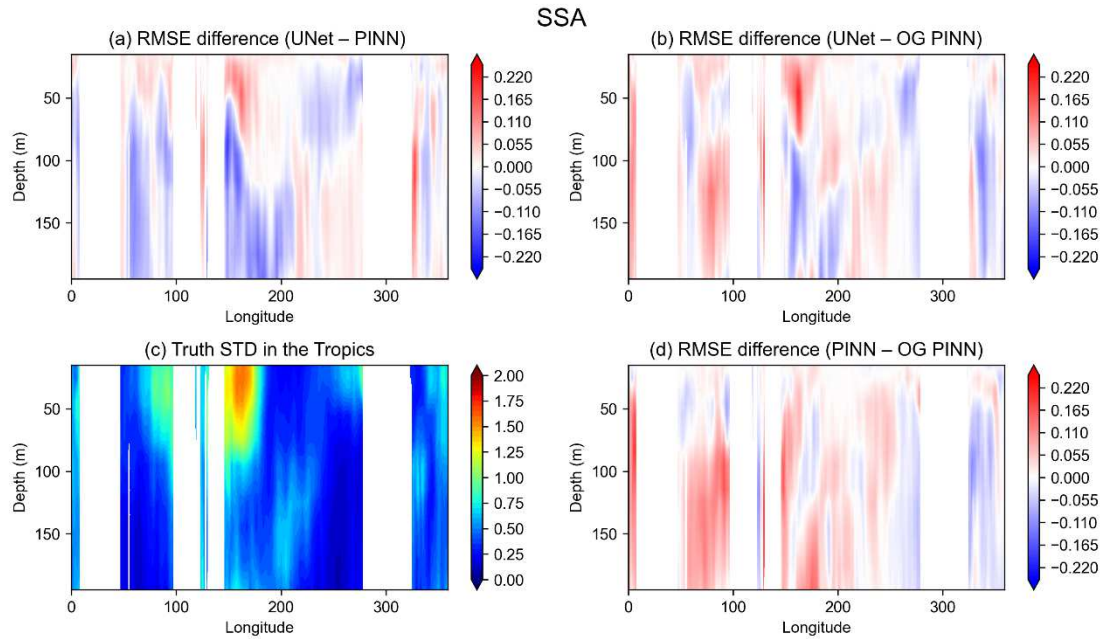


Figure 14 Comparison of SSA reconstruction RMSE among different models in the equatorial band. (a) UNet minus standard PINN; (b) UNet minus OG-PINN; (c) SODA SSA standard deviation (STD) over the equator, indicating the spatial pattern of natural variability; (d) PINN minus OG-PINN. All subplots are longitude–depth sections (5°S–5°N average).

The STD of SDA closely resembles that of STA, except for an additional high-variance region in the equatorial western Pacific (Fig. 15c). This indicates that SDA in the thermocline is primarily controlled by STA, whereas near the western Pacific warm pool, it is more strongly influenced by SSA.

OG-PINN effectively reduces SDA reconstruction errors across most high-STD regions, particularly in the upper eastern Pacific and throughout the entire water column of the western Pacific, and performs poorly only in the western Indian Ocean. In contrast, standard PINN performs poorly in several key regions, including the upper

eastern Pacific (shallower than 50 m) and the deep western Pacific (deeper than 150 m), where its RMSE even exceeds that of U-Net.

A direct comparison between OG-PINN and standard PINN further reveals that OG-PINN achieves lower RMSE in nearly all high-STD regions of SDA. However, slight RMSE increases are observed in the western Indian Ocean, the upper Pacific (shallower than 50 m), and the deep Atlantic (deeper than 100 m). This suggests that the observation-guided mechanism provides limited improvement for SDA reconstruction in these areas, likely because the physical constraints in OG-PINN are not directly applied to the density variable itself (as noted in Equation 5).

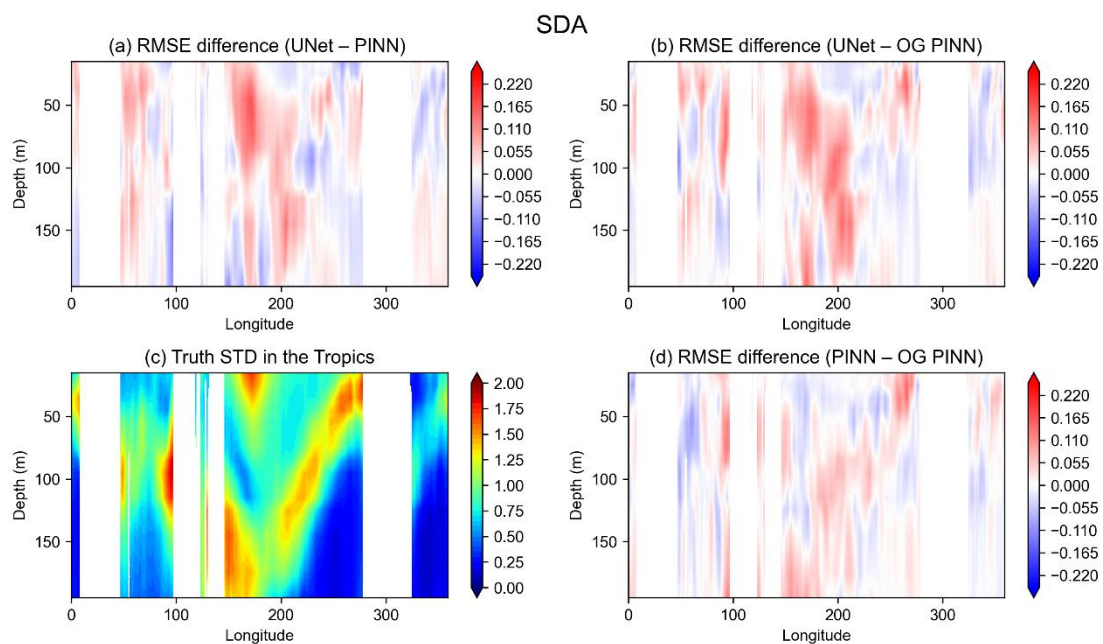


Figure 15 Comparison of SDA reconstruction RMSE among different models in the equatorial band. (a) UNet minus standard PINN; (b) UNet minus OG-PINN; (c) SODA SDA standard deviation (STD) over the equator, indicating the spatial pattern of natural variability; (d) PINN minus OG-PINN. All subplots are longitude–depth sections (5°S–5°N average).

As the dynamical and thermal engine of the global climate system, the equatorial region plays a pivotal role in driving climate variability. Accurate reconstruction of its T-S structure is essential for understanding and predicting global climate variability.

Results in this section demonstrate that, despite incorporating physical constraints, the standard PINN often yields spurious physical consistency in dynamically complex regions. As a result, its reconstruction performance falls below that of purely data-driven models. In contrast, OG-PINN, through its observation-guided mechanism, effectively balances the competition between data fitting and physical constraints. It significantly improves the reconstruction accuracy of STA, SSA and SDA across the thermocline and along wave propagation pathways, with particularly notable improvements in the eastern Pacific. Overall, OG-PINN exhibits superior robustness and physical consistency in the equator’s complex dynamical environment.

860

861 **5. Conclusions and Discussion**

862 This study proposes an Observation-Guided Physics-Informed Neural Network
863 (OG-PINN) to efficiently reconstruct three-dimensional fields of subsurface ocean
864 temperature, salinity, and density while enhancing their physical consistency. The core
865 innovation is to embed the seawater equation of state as a physical constraint within the
866 network and integrating observational information directly into the physics loss term.
867 This design effectively mitigates the optimization conflict between data and physics
868 constraints present in traditional PINNs. This design steers the physical constraint
869 toward convergence on the true ocean state and strengthens physical consistency among
870 different variables.

871 Systematic evaluations reveal that OG-PINN consistently outperforms baseline
872 models across diverse spatial scales. For STA, high-correlation reconstructions (>0.6)
873 are achieved throughout the upper 200 m, with the most pronounced improvements
874 occurring near the thermocline (50–115 m), where RMSE is significantly reduced. SSA,
875 governed by multiscale hydrological processes and inherently less predictable, still
876 benefits from stable quality improvements by OG-PINN in the upper 100 m. SDA also
877 achieves high-correlation reconstruction (>0.6) within 200 m, with peak performance
878 gains centered around 50 m depth.

879 Notably, the performance advantage of OG-PINN over standard PINN exhibits
880 strong regional dependence. STA improvements are concentrated in the central Pacific,
881 cold tongue, and equatorial eastern Atlantic. SSA improvements are most evident in the
882 western Indian Ocean upwelling region. At 95 m depth, the maximum correlation
883 increases reach 0.30 for STA and 0.58 for SSA, respectively.

884 Importantly, in key equatorial climate regions (e.g., the eastern Pacific cold tongue
885 and the western Pacific warm pool), OG-PINN captures thermocline structures
886 significantly better than U-Net and the standard PINN, providing more informative
887 initial fields for climate modes such as ENSO.

888 Further analysis reveals that all physics-constrained models deteriorate in strong
889 upwelling zones, with performance losses intensifying with depth. This pattern aligns
890 closely with the climatological vertical velocity (w) field. In these regions, subsurface
891 water masses are dominated by local vertical transport and are strongly decoupled from
892 surface signals, rendering static thermodynamic constraints inadequate to represent true
893 dynamics. Standard PINN, lacking ground-truth ocean state information, often
894 converges to mathematically consistent yet physically unrealistic solutions. By contrast,
895 OG-PINN's observation-guided mechanism both strengthens the alignment between
896 physical constraints and the true state and identifies areas where physical priors lose
897 applicability. Through a spatially adaptive weighting strategy, it dynamically
898 downweights constraints in those areas, thereby maintaining robustness while avoiding
899 over-regularization.

900 Nevertheless, this study also highlights current limitations: in deep layers of core
901 upwelling regions, exclusive reliance on surface observations and a static equation of
902 state is insufficient to accurately reconstruct subsurface T-S fields. Here, OG-PINN's

advantage rapidly diminishes and can even fall below that of purely data-driven models. This underscores a fundamental shortcoming of static thermodynamic priors in dynamically active zones. Future work should incorporate dynamic constraints, such as vertical velocity, vorticity, mixing rates, or simplified dynamical equations, to develop a next-generation PINN framework featuring coupled thermodynamic–dynamic constraints.

In summary, OG-PINN validates the effectiveness and interpretability of the “observation-guided + adaptive physics constraint” paradigm for subsurface ocean reconstruction. It provides a promising new pathway toward high-temporal-resolution, physically consistent ocean data assimilation and reconstruction systems. The high-quality 3D T-S fields generated by OG-PINN can directly support climate model initialization, potentially enhancing the predictability of key climate modes such as ENSO and IOD, offering significant scientific value and practical applications.

CRedit authorship contribution statement:

Yao Xiao: Conceptualization, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing, Validation; Youmin Tang: Conceptualization, Formal analysis, Supervision, Writing - review & editing; Yi Li: Investigation, Writing – review & editing.

Data availability

The reanalysis datasets were downloaded from the Simple Ocean Data Assimilation (SODA) reanalysis Datasets (<http://apdrc.soest.hawaii.edu/las/v6/dataset?catitem=4782>).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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